DATA RECORD

Titanium Model A V6

Training dataset lineage, field purpose and use in prediction

ROWS	COLUMNS	COVERAGE	PRODUCTS
396	**134**	**Jan 2010 to Jun 2026**	**HS 810820 and 810890**

Purpose

This PDF records exactly what the selected V6 training CSV contains, where each data family originates, and how it supports the direct 3, 6 and 12-month GBP/kg forecasts. The file is a reproducible training snapshot, not a raw-source export. It combines source observations, engineered model inputs, future training labels, publication-timing controls and audit fields.

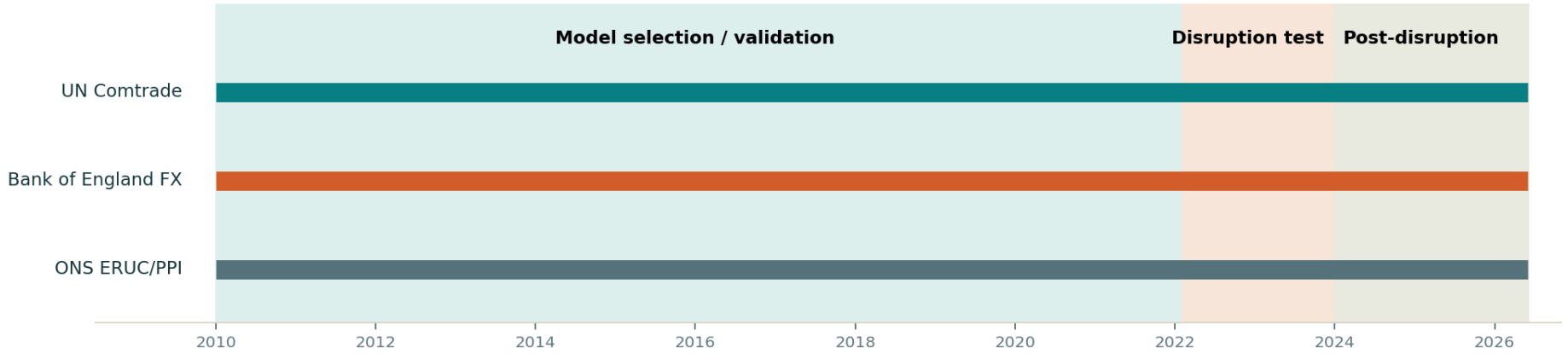
Important interpretation

The observed price is an import unit value: UK import value divided by imported net weight. It reflects product mix, supplier mix, contract timing and data quality as well as underlying titanium-market conditions. It must not be described as a spot-market quotation.

41	Selected V6 model inputs	**134**	Total columns including targets, provenance and controls

1. Source lineage

Source	Supplied file(s)	Fields contributed	Use
UN Comtrade	17 annual UK monthly-import CSV exports	USD trade value, net weight, partner countries, HS identity and source-quality metadata	Creates GBP/kg target, volume history, supplier mix and concentration features
Bank of England	Bank of England Database.csv	Monthly average USD per GBP and derived FX changes	Converts USD trade value to GBP and captures sterling movement
ONS	ppi.csv, ERUC series	C24 basic-metals import-price index, 2015=100, and lagged changes	Broad non-ferrous imported-cost signal; not titanium-specific
Preparation pipeline	Deterministic derived fields	Lags, volatility, supplier structure, seasonality, quality flags and direct future labels	Transforms time-ordered source data into leak-controlled training examples



Time boundary

Each observation is one month and one HS code. Model selection and blocked validation end in January 2022. February 2022 through December 2023 is the frozen disruption test. January 2024 onward is post-disruption monitoring. Availability dates prevent source values or labels from entering a simulated forecast before they would have been published.

Explicit exclusions

The V6 snapshot contains no USGS titanium-sponge proxy and no World Bank aluminium or nickel price features. HS 810830 titanium waste and scrap is also excluded. These exclusions keep the selected dataset focused on titanium trade, FX and the documented broad ERUC/PPI proxy.

2. From source records to GBP/kg

Step	Calculation	Why it is needed
Weight selection	Use positive World net weight; otherwise sum partner-country weights and set reconstruction flag	Preserves months where the aggregate World weight is missing or zero
GBP value	trade_value_gbp = trade_value_usd / usd_per_gbp_monthly_average	Places import expenditure in the model's required currency
GBP/kg unit value	import_unit_value_gbp_per_kg = trade_value_gbp / net_weight_kg	Creates the observed monthly target-price proxy
Direct H3/H6/H12 label	ln(price at t+h) - ln(price at t)	Learns each horizon directly rather than recursively chaining H1 predictions

Two separate product series

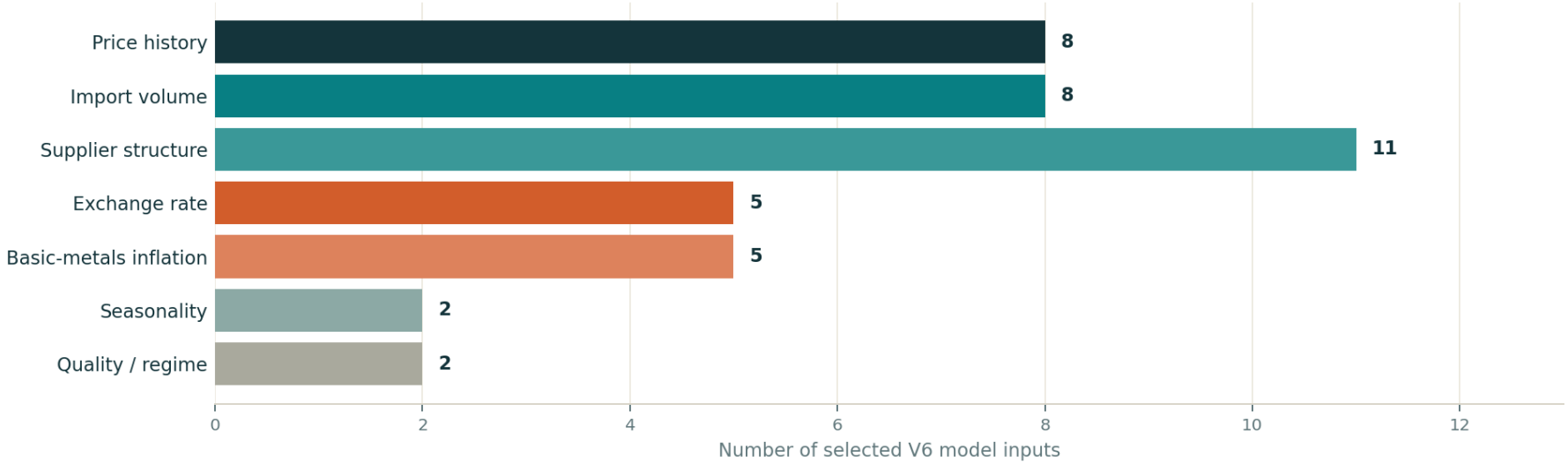
HS code	Product	Rows	Reason for separation
810820	Titanium unwrought and powders	198	Different product form, price level and supplier behaviour
810890	Titanium bars, sheet, forgings and other articles	198	Different product form, price level and supplier behaviour

Data-quality handling

Flagged observations are retained for traceability rather than silently overwritten. The training pipeline uses explicit weight-reconstruction, estimated-weight, low-volume, price-shock and structural-change indicators. Eligibility flags and assumed publication dates control whether a row can train a given horizon.

3. The 41 predictors used by V6

Only the following manifest-recorded columns are supplied to the selected models. Future targets, future prices, filenames and evaluation labels are not model inputs.



Feature	Group	Source	Training/prediction role
price_log_change_gbp_1m	Price history	Derived: Comtrade + FX	Learns momentum, mean reversion and recent instability in the GBP/kg target.
price_log_change_gbp_2m	Price history	Derived: Comtrade + FX	Learns momentum, mean reversion and recent instability in the GBP/kg target.
price_log_change_gbp_3m	Price history	Derived: Comtrade + FX	Learns momentum, mean reversion and recent instability in the GBP/kg target.
price_log_change_gbp_6m	Price history	Derived: Comtrade + FX	Learns momentum, mean reversion and recent instability in the GBP/kg target.
price_log_change_gbp_12m	Price history	Derived: Comtrade + FX	Learns momentum, mean reversion and recent instability in the GBP/kg target.
price_volatility_6m	Price history	Derived from trade/FX	Learns momentum, mean reversion and recent instability in the GBP/kg target.
price_volatility_12m	Price history	Derived from trade/FX	Learns momentum, mean reversion and recent instability in the GBP/kg target.
quantity_log_change_1m	Import volume	Derived from trade/FX	Captures demand/supply intensity and warns when low volume makes unit values unstable.
quantity_log_change_3m	Import volume	Derived from trade/FX	Captures demand/supply intensity and warns when low volume makes unit values unstable.
quantity_log_change_12m	Import volume	Derived from trade/FX	Captures demand/supply intensity and warns when low volume makes unit values unstable.
quantity_volatility_6m	Import volume	Derived from trade/FX	Captures demand/supply intensity and warns when low volume makes unit values unstable.
quantity_volatility_12m	Import volume	Derived from trade/FX	Captures demand/supply intensity and warns when low volume makes unit values unstable.

Feature	Group	Source	Training/prediction role
log_net_weight_kg	Import volume	Derived feature	Captures demand/supply intensity and warns when low volume makes unit values unstable.
volume_ratio_to_prior_12m_median	Import volume	Derived feature	Captures demand/supply intensity and warns when low volume makes unit values unstable.
low_volume_history_flag	Import volume	Derived feature	Captures demand/supply intensity and warns when low volume makes unit values unstable.
supplier_share_usa	Supplier structure	Derived from UN Comtrade	Represents sourcing mix, concentration and shifts among major supplying countries.
supplier_share_japan	Supplier structure	Derived from UN Comtrade	Represents sourcing mix, concentration and shifts among major supplying countries.
supplier_share_france	Supplier structure	Derived from UN Comtrade	Represents sourcing mix, concentration and shifts among major supplying countries.
supplier_share_russia	Supplier structure	Derived from UN Comtrade	Represents sourcing mix, concentration and shifts among major supplying countries.
supplier_share_china	Supplier structure	Derived from UN Comtrade	Represents sourcing mix, concentration and shifts among major supplying countries.
supplier_share_other	Supplier structure	Derived from UN Comtrade	Represents sourcing mix, concentration and shifts among major supplying countries.
supplier_hhi_weight	Supplier structure	Derived from UN Comtrade	Represents sourcing mix, concentration and shifts among major supplying countries.
top_supplier_share_weight	Supplier structure	Derived from UN Comtrade	Represents sourcing mix, concentration and shifts among major supplying countries.
supplier_concentration_change_3m	Supplier structure	Derived from UN Comtrade	Represents sourcing mix, concentration and shifts among major supplying countries.
top_supplier_share_change_3m	Supplier structure	Derived from UN Comtrade	Represents sourcing mix, concentration and shifts among major supplying countries.
log_partner_count	Supplier structure	Derived feature	Represents sourcing mix, concentration and shifts among major supplying countries.
fx_log_change_1m	Exchange rate	Bank of England	Separates sterling movement from changes in the underlying USD-denominated trade value.
fx_log_change_3m	Exchange rate	Bank of England	Separates sterling movement from changes in the underlying USD-denominated trade value.
fx_log_change_6m	Exchange rate	Bank of England	Separates sterling movement from changes in the underlying USD-denominated trade value.
fx_log_change_12m	Exchange rate	Bank of England	Separates sterling movement from changes in the underlying USD-denominated trade value.
fx_volatility_6m	Exchange rate	Bank of England	Separates sterling movement from changes in the underlying USD-denominated trade value.
eruc_log_change_1m	Basic-metals inflation	ONS ERUC/PPI	Provides a broad imported-basic-metals cost signal; it is not titanium-specific.
eruc_log_change_3m	Basic-metals inflation	ONS ERUC/PPI	Provides a broad imported-basic-metals cost signal; it is not titanium-specific.
eruc_log_change_6m	Basic-metals inflation	ONS ERUC/PPI	Provides a broad imported-basic-metals cost signal; it is not titanium-specific.
eruc_log_change_12m	Basic-metals inflation	ONS ERUC/PPI	Provides a broad imported-basic-metals cost signal; it is not titanium-specific.
eruc_volatility_6m	Basic-metals inflation	ONS ERUC/PPI	Provides a broad imported-basic-metals cost signal; it is not titanium-specific.
month_sin	Seasonality	Derived feature	Allows recurring month-of-year effects without treating December and January as far apart.
month_cos	Seasonality	Derived feature	Allows recurring month-of-year effects without treating December and January as far apart.
price_shock_history_flag	Price history	Derived from trade/FX	Learns momentum, mean reversion and recent instability in the GBP/kg target.
world_weight_estimated_by_un_flag	Quality / regime	UN Comtrade	Marks shocks, estimated weights and structural collection changes that affect reliability.
eu_collection_change_flag	Quality / regime	Derived feature	Marks shocks, estimated weights and structural collection changes that affect reliability.

4. Targets, controls and leakage boundary

Field family	Purpose	Allowed use
target_log_change_gbp_per_kg_h3/h6/h12	Direct future log-change labels for each horizon	Training outcome only; never a predictor
target_unit_value_gbp_per_kg_h3/h6/h12	Observed future GBP/kg values for audit and evaluation	Target/audit only
target_month_gbp_h3/h6/h12	Identifies the future month associated with each label	Audit and reporting
target_data_available_date_assumed_h*	Models when the outcome would become observable	Training-window eligibility
model_a_training_eligible_h*	Combines feature completeness, source completeness, quality and target availability	Row selection, not a predictive signal
evaluation_partition	Separates selection, stress-test and monitoring periods	Evaluation control only

Leakage controls

- Source publication dates are compared with the simulated forecast date.

- A training row is used only after its H3, H6 or H12 outcome is assumed available.

- Targets and future prices remain in the CSV for auditability but are excluded from the manifest feature list.

- Validation is blocked in time; the 2022-2023 disruption period does not select the model family.

- Each HS product and horizon is modelled separately, avoiding price-level averaging across unlike titanium forms.

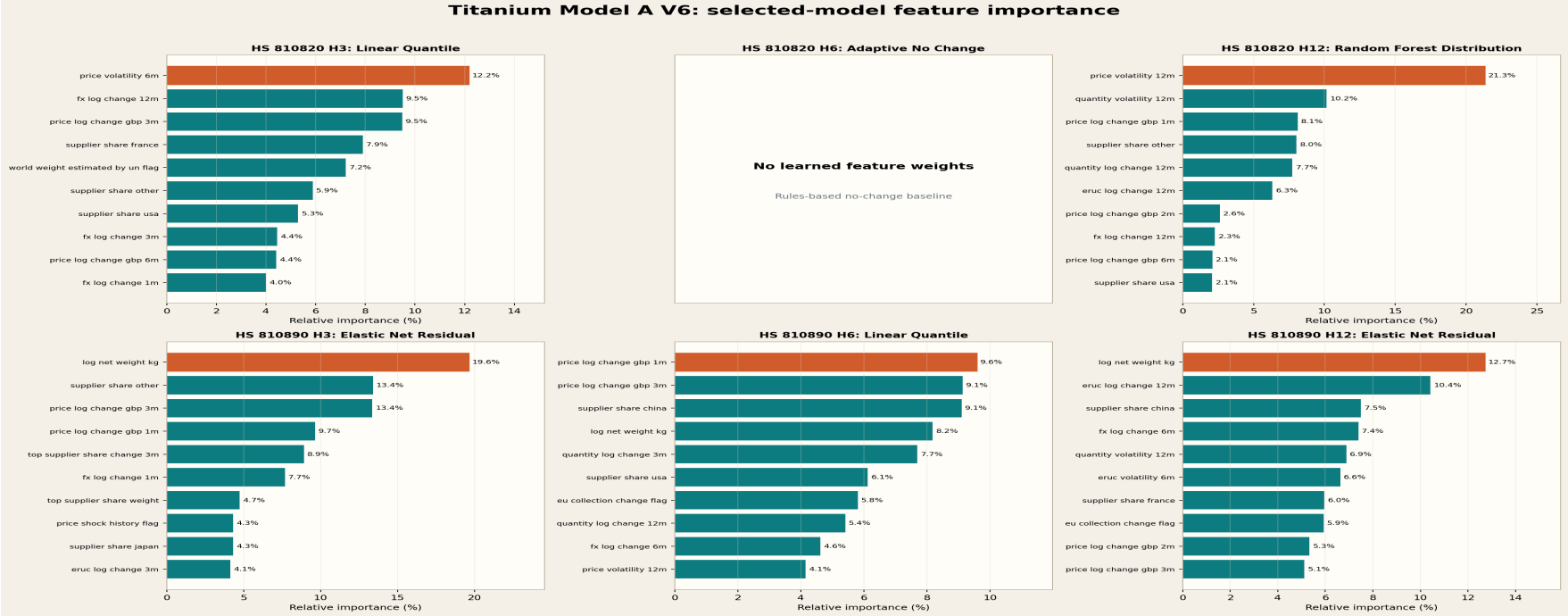
Availability at the end of the dataset

The latest rows naturally have missing future H3, H6 or H12 labels because those target months have not yet occurred or were not yet assumed published. This is expected and is not source-data missingness. Such rows can be forecast origins but cannot yet be completed training examples for the affected horizon.

5. How the data supports prediction

Question	Data used	What the model can learn
Is price momentum changing?	Price log changes and rolling volatility	Recent direction, persistence, mean reversion and instability
Is the unit value based on normal trade volume?	Net weight, quantity changes, volume ratio and low-volume flag	Whether a price movement coincides with a demand/supply shift or thin-volume noise
Has the sourcing mix changed?	Country shares, HHI, top-supplier share and their changes	Supplier concentration, substitution and exposure to dominant partners
Did sterling move?	Bank of England FX changes and volatility	Currency contribution to GBP-denominated import unit values
Are imported basic-metal costs shifting?	ERUC/PPI changes and volatility	Broad non-ferrous import-cost pressure; treated as a proxy, not titanium causality
Are there recurring or structural effects?	Month sine/cosine, EU collection-change and quality flags	Seasonality, regime changes and data-reliability conditions

Selected-model feature influence



What the dataset does not establish

Feature importance and predictive association do not establish causality. ERUC is broad, UN Comtrade unit values reflect product mix, and supplier shares may proxy changes in specification or contracting. The resulting forecast should support procurement planning together with domain review, not replace it.

6. Dataset controls and reproducibility

Control	Observed in V6 snapshot	Interpretation
Rows / columns	396 / 134	One row per HS code and month
Date range	2010-01 to 2026-06	Continuous monthly modelling span
Selected features	41	Must match model_manifest.json
Rows with at least one unavailable selected feature	24	Expected mainly in early lag windows; eligibility controls fitting
Excluded proxy columns detected	0	No USGS or World Bank cross-metal inputs remain

Reproduction files

The training snapshot is consumed by code/train_model_a_production_v6_titanium_core_corrected.py. Selected feature names, model files, horizons and calibration metadata are recorded in results/latest/model_manifest.json. RUN_TRAINING_V6.cmd reproduces training and evaluation from the isolated V6 package.

Known limitations

- ERUC/PPI is a broad C24 basic-metals import-price proxy, not a titanium benchmark.

- The assumed 42-day trade, 7-day FX and 31-day ERUC publication lags should be validated against production release calendars.

- Import unit values may change because of product grade and contract mix even if the underlying market is unchanged.

- The 2024-2025 assessment is useful but not pristine because this dataset was previously explored.

- A genuinely prospective holdout is still required before production approval.

Appendix A. Full 134-column data dictionary

Every column in the delivered CSV is listed below. 'Model input' means the field appears in the saved V6 manifest. Other columns remain for conversion, row selection, evaluation, quality control or auditability.

Column	Source	Role	Definition	Missing
date	UN Comtrade	Identity / split control	Calendar month represented by the row, stored as the first day of the month.	0 (0.0%)
period	UN Comtrade	Identity / split control	UN Comtrade monthly period identifier.	0 (0.0%)
year	UN Comtrade	Audit / supporting field	Calendar year extracted from the monthly period.	0 (0.0%)
month	UN Comtrade	Audit / supporting field	Calendar month number, 1 to 12.	0 (0.0%)
hs_code	UN Comtrade	Identity / split control	Titanium product category. V6 uses 810820 and 810890 separately.	0 (0.0%)
hs_product	UN Comtrade	Identity / split control	Readable product name for the HS code.	0 (0.0%)
trade_data_available_date_assumed	UN Comtrade	Identity / split control	Conservative month-end plus 42-day date used to prevent publication-timing leakage.	0 (0.0%)
feature_set_id	UN Comtrade	Audit / supporting field	Version label identifying the prepared feature specification.	0 (0.0%)
trade_value_usd	UN Comtrade	Audit / supporting field	UK monthly import value reported by UN Comtrade, in USD.	0 (0.0%)
trade_value_gbp	Derived: Comtrade + FX	Audit / supporting field	Trade value converted to GBP using the monthly USD-per-GBP rate.	0 (0.0%)
net_weight_kg	UN Comtrade	Audit / supporting field	Selected monthly import weight used in unit-value calculations.	0 (0.0%)
import_unit_value_usd_per_kg	UN Comtrade	Audit / supporting field	Trade value USD divided by selected net weight. This is an import unit value, not a quoted market price.	0 (0.0%)
usd_per_gbp_monthly_average	Bank of England	Audit / supporting field	Bank of England monthly average: US dollars per one pound sterling.	0 (0.0%)
import_unit_value_gbp_per_kg	Derived: Comtrade + FX	Audit / supporting field	Primary observed price proxy: GBP trade value divided by net weight in kg.	0 (0.0%)
log_unit_value_gbp	Derived: Comtrade + FX	Audit / supporting field	Natural logarithm of the GBP/kg import unit value.	0 (0.0%)
fx_period	Bank of England	Audit / supporting field	Month matched to the Bank of England FX observation.	0 (0.0%)
fx_available_date_assumed	Bank of England	Audit / supporting field	Month-end plus seven days, used as the assumed FX availability date.	0 (0.0%)
fx_known_by_forecast_date	Bank of England	Quality / timing control	True when the FX observation would have been available at the simulated forecast date.	0 (0.0%)
fx_log_change_1m	Bank of England	Model input	Log change in USD per GBP over 1 month(s); captures currency movement.	2 (0.5%)
fx_log_change_3m	Bank of England	Model input	Log change in USD per GBP over 3 month(s); captures currency movement.	6 (1.5%)
fx_log_change_6m	Bank of England	Model input	Log change in USD per GBP over 6 month(s); captures currency movement.	12 (3.0%)
fx_log_change_12m	Bank of England	Model input	Log change in USD per GBP over 12 month(s); captures currency movement.	24 (6.1%)
eruc_series_id	ONS ERUC/PPI	Audit / supporting field	ONS time-series identifier ERUC.	0 (0.0%)
eruc_title	ONS ERUC/PPI	Audit / supporting field	ONS description of the broad basic-metals import-price index.	0 (0.0%)
eruc_period	ONS ERUC/PPI	Audit / supporting field	Month of the ERUC/PPI observation.	0 (0.0%)

Column	Source	Role	Definition	Missing
eruc_index_2015_100	ONS ERUC/PPI	Audit / supporting field	ONS import producer-price index for C24 basic metals, with 2015=100.	0 (0.0%)
eruc_available_date_assumed	ONS ERUC/PPI	Audit / supporting field	Month-end plus 31 days, used to avoid publication-timing leakage.	0 (0.0%)
eruc_known_by_forecast_date	ONS ERUC/PPI	Quality / timing control	True when the ERUC value would have been available at forecast time.	0 (0.0%)
eruc_log_change_1m	ONS ERUC/PPI	Model input	Log change in the ONS ERUC basic-metals import-price index over 1 month(s).	0 (0.0%)
eruc_log_change_3m	ONS ERUC/PPI	Model input	Log change in the ONS ERUC basic-metals import-price index over 3 month(s).	0 (0.0%)
eruc_log_change_6m	ONS ERUC/PPI	Model input	Log change in the ONS ERUC basic-metals import-price index over 6 month(s).	0 (0.0%)
eruc_log_change_12m	ONS ERUC/PPI	Model input	Log change in the ONS ERUC basic-metals import-price index over 12 month(s).	0 (0.0%)
price_log_change_gbp_1m	Derived: Comtrade + FX	Model input	Log change in GBP/kg import unit value versus 1 month(s) earlier.	2 (0.5%)
price_log_change_gbp_2m	Derived: Comtrade + FX	Model input	Log change in GBP/kg import unit value versus 2 month(s) earlier.	4 (1.0%)
price_log_change_gbp_3m	Derived: Comtrade + FX	Model input	Log change in GBP/kg import unit value versus 3 month(s) earlier.	6 (1.5%)
price_log_change_gbp_6m	Derived: Comtrade + FX	Model input	Log change in GBP/kg import unit value versus 6 month(s) earlier.	12 (3.0%)
price_log_change_gbp_12m	Derived: Comtrade + FX	Model input	Log change in GBP/kg import unit value versus 12 month(s) earlier.	24 (6.1%)
quantity_log_change_1m	Derived from trade/FX	Model input	Log change in imported net weight versus 1 month(s) earlier.	2 (0.5%)
quantity_log_change_3m	Derived from trade/FX	Model input	Log change in imported net weight versus 3 month(s) earlier.	6 (1.5%)
quantity_log_change_12m	Derived from trade/FX	Model input	Log change in imported net weight versus 12 month(s) earlier.	24 (6.1%)
supplier_share_usa	Derived from UN Comtrade	Model input	Share of monthly import weight supplied by Usa.	0 (0.0%)
supplier_share_japan	Derived from UN Comtrade	Model input	Share of monthly import weight supplied by Japan.	0 (0.0%)
supplier_share_france	Derived from UN Comtrade	Model input	Share of monthly import weight supplied by France.	0 (0.0%)
supplier_share_russia	Derived from UN Comtrade	Model input	Share of monthly import weight supplied by Russia.	0 (0.0%)
supplier_share_china	Derived from UN Comtrade	Model input	Share of monthly import weight supplied by China.	0 (0.0%)
supplier_share_other	Derived from UN Comtrade	Model input	Share of monthly import weight supplied by Other.	0 (0.0%)
eu_collection_change_flag	Derived feature	Model input	Indicator for the January 2021 UK/EU trade-data collection change.	0 (0.0%)
model_a_selected_feature_complete	Derived control	Audit / supporting field	True when every selected predictor is populated.	0 (0.0%)
model_a_source_inputs_complete	Derived control	Audit / supporting field	True when required trade, FX and ERUC source inputs are present.	0 (0.0%)
target_month_gbp_h3	Derived: Comtrade + FX	Target / audit only	Future target month 3 months after the forecast origin. Audit field only.	6 (1.5%)
target_unit_value_gbp_per_kg_h3	Derived: Comtrade + FX	Target / audit only	Observed future GBP/kg unit value at H3. Target/audit field, never an input.	6 (1.5%)
target_log_change_gbp_per_kg_h3	Derived: Comtrade + FX	Target / audit only	Direct H3 training label: log future/current GBP/kg unit value. Never an input.	6 (1.5%)
target_data_available_date_assumed_h3	Derived from trade/FX	Audit / supporting field	Assumed publication date of the H3 outcome, used to enforce target availability.	6 (1.5%)
model_a_training_eligible_h3	Derived control	Row-selection control	Final V6 training eligibility flag for horizon H3.	0 (0.0%)

Column	Source	Role	Definition	Missing
target_month_gbp_h6	Derived: Comtrade + FX	Target / audit only	Future target month 6 months after the forecast origin. Audit field only.	12 (3.0%)
target_unit_value_gbp_per_kg_h6	Derived: Comtrade + FX	Target / audit only	Observed future GBP/kg unit value at H6. Target/audit field, never an input.	12 (3.0%)
target_log_change_gbp_per_kg_h6	Derived: Comtrade + FX	Target / audit only	Direct H6 training label: log future/current GBP/kg unit value. Never an input.	12 (3.0%)
target_data_available_date_assumed_h6	Derived from trade/FX	Audit / supporting field	Assumed publication date of the H6 outcome, used to enforce target availability.	12 (3.0%)
model_a_training_eligible_h6	Derived control	Row-selection control	Final V6 training eligibility flag for horizon H6.	0 (0.0%)
target_month_gbp_h12	Derived: Comtrade + FX	Target / audit only	Future target month 12 months after the forecast origin. Audit field only.	24 (6.1%)
target_unit_value_gbp_per_kg_h12	Derived: Comtrade + FX	Target / audit only	Observed future GBP/kg unit value at H12. Target/audit field, never an input.	24 (6.1%)
target_log_change_gbp_per_kg_h12	Derived: Comtrade + FX	Target / audit only	Direct H12 training label: log future/current GBP/kg unit value. Never an input.	24 (6.1%)
target_data_available_date_assumed_h12	Derived from trade/FX	Audit / supporting field	Assumed publication date of the H12 outcome, used to enforce target availability.	24 (6.1%)
model_a_training_eligible_h12	Derived control	Row-selection control	Final V6 training eligibility flag for horizon H12.	0 (0.0%)
source_product_description	UN Comtrade	Audit / supporting field	Commodity description supplied in the UN Comtrade export.	0 (0.0%)
classification_code	UN Comtrade	Audit / supporting field	Trade classification/version recorded by UN Comtrade.	0 (0.0%)
source_file	UN Comtrade	Audit / supporting field	Annual UN Comtrade CSV from which the monthly record originated.	0 (0.0%)
source_extraction_date	UN Comtrade	Audit / supporting field	Date the UN Comtrade source export was obtained.	0 (0.0%)
month_end_date	UN Comtrade	Audit / supporting field	Last calendar date of the observation month.	0 (0.0%)
evaluation_partition	UN Comtrade	Identity / split control	Time-ordered selection, disruption-test or post-disruption partition.	0 (0.0%)
world_net_weight_kg_original	UN Comtrade	Audit / supporting field	Net weight on the World aggregate record before reconstruction.	12 (3.0%)
partner_weight_sum_kg	UN Comtrade	Audit / supporting field	Sum of partner-country net weights for the same HS code and month.	0 (0.0%)
weight_source	UN Comtrade	Audit / supporting field	Indicates whether weight came from the World row or partner reconstruction.	0 (0.0%)
weight_reconstructed_flag	UN Comtrade	Quality / timing control	1 when missing/zero World weight was replaced by summed partner weights.	0 (0.0%)
world_weight_estimated_by_un_flag	UN Comtrade	Model input	UN Comtrade flag indicating that the World weight was estimated.	0 (0.0%)
partner_value_sum_usd	UN Comtrade	Audit / supporting field	Sum of partner-country import values for reconciliation.	0 (0.0%)
world_value_partner_difference_pct	UN Comtrade	Audit / supporting field	Percentage difference between World value and summed partner values.	0 (0.0%)
partner_count	UN Comtrade	Audit / supporting field	Number of partner countries with trade in the month.	0 (0.0%)
log_unit_value	UN Comtrade	Audit / supporting field	Natural logarithm of the USD/kg import unit value.	0 (0.0%)
unit_value_robust_z	UN Comtrade	Audit / supporting field	Robust outlier score for the monthly unit value.	0 (0.0%)
unit_value_outlier_flag	UN Comtrade	Quality / timing control	Flags extreme unit values without silently replacing them.	0 (0.0%)
partner_anomaly_count	UN Comtrade	Audit / supporting field	Count of suspicious partner-level value/weight combinations in the month.	0 (0.0%)
partner_anomaly_max_weight_multiple	UN Comtrade	Audit / supporting field	Largest detected partner-weight anomaly multiple.	383 (96.7%)

Column	Source	Role	Definition	Missing
data_quality_status	UN Comtrade	Quality / timing control	Summary label for source and unit-value quality checks.	0 (0.0%)
price_log_change_1m	Derived from trade/FX	Audit / supporting field	Log change in USD/kg import unit value versus 1 month(s) earlier.	2 (0.5%)
price_log_change_2m	Derived from trade/FX	Audit / supporting field	Log change in USD/kg import unit value versus 2 month(s) earlier.	4 (1.0%)
price_log_change_3m	Derived from trade/FX	Audit / supporting field	Log change in USD/kg import unit value versus 3 month(s) earlier.	6 (1.5%)
price_log_change_6m	Derived from trade/FX	Audit / supporting field	Log change in USD/kg import unit value versus 6 month(s) earlier.	12 (3.0%)
price_log_change_12m	Derived from trade/FX	Audit / supporting field	Log change in USD/kg import unit value versus 12 month(s) earlier.	24 (6.1%)
supplier_share_sum_check	Derived from UN Comtrade	Quality / timing control	Check that supplier weight shares sum to approximately one.	0 (0.0%)
supplier_hhi_weight	Derived from UN Comtrade	Model input	Herfindahl-Hirschman concentration index of supplier weight shares.	0 (0.0%)
top_supplier_name	Derived from UN Comtrade	Audit / supporting field	Partner country with the largest monthly import-weight share.	0 (0.0%)
top_supplier_share_weight	Derived from UN Comtrade	Model input	Weight share supplied by the largest partner country.	0 (0.0%)
month_sin	Derived feature	Model input	Sine encoding of calendar month for smooth annual seasonality.	0 (0.0%)
month_cos	Derived feature	Model input	Cosine encoding of calendar month for smooth annual seasonality.	0 (0.0%)
latest_year_partial_flag	Derived feature	Quality / timing control	Indicates that the latest calendar year is incomplete.	0 (0.0%)
no_change_baseline_log_change	Derived feature	Audit / supporting field	Zero log-change benchmark used by the no-change baseline.	0 (0.0%)
target_month_h3	Derived from trade/FX	Target / audit only	Future target month 3 months after the forecast origin. Audit field only.	6 (1.5%)
target_unit_value_usd_per_kg_h3	Derived from trade/FX	Target / audit only	Observed future USD/kg unit value at H3. Target/audit field, never an input.	6 (1.5%)
target_log_change_usd_per_kg_h3	Derived from trade/FX	Target / audit only	Direct H3 training label: log future/current USD/kg unit value. Never an input.	6 (1.5%)
target_quality_flag_h3	Derived control	Quality / timing control	Quality status for the H3 target observation.	0 (0.0%)
trade_core_eligible_h3	UN Comtrade	Row-selection control	Trade-core eligibility flag for direct H3 modelling.	0 (0.0%)
target_month_h6	Derived from trade/FX	Target / audit only	Future target month 6 months after the forecast origin. Audit field only.	12 (3.0%)
target_unit_value_usd_per_kg_h6	Derived from trade/FX	Target / audit only	Observed future USD/kg unit value at H6. Target/audit field, never an input.	12 (3.0%)
target_log_change_usd_per_kg_h6	Derived from trade/FX	Target / audit only	Direct H6 training label: log future/current USD/kg unit value. Never an input.	12 (3.0%)
target_quality_flag_h6	Derived control	Quality / timing control	Quality status for the H6 target observation.	0 (0.0%)
trade_core_eligible_h6	UN Comtrade	Row-selection control	Trade-core eligibility flag for direct H6 modelling.	0 (0.0%)
target_month_h12	Derived from trade/FX	Target / audit only	Future target month 12 months after the forecast origin. Audit field only.	24 (6.1%)
target_unit_value_usd_per_kg_h12	Derived from trade/FX	Target / audit only	Observed future USD/kg unit value at H12. Target/audit field, never an input.	24 (6.1%)
target_log_change_usd_per_kg_h12	Derived from trade/FX	Target / audit only	Direct H12 training label: log future/current USD/kg unit value. Never an input.	24 (6.1%)
target_quality_flag_h12	Derived control	Quality / timing control	Quality status for the H12 target observation.	0 (0.0%)
trade_core_eligible_h12	UN Comtrade	Row-selection control	Trade-core eligibility flag for direct H12 modelling.	0 (0.0%)

Column	Source	Role	Definition	Missing
eruc_source_release_date	ONS ERUC/PPI	Audit / supporting field	Release date recorded in the downloaded ONS source file.	0 (0.0%)
eruc_source_extraction_date	ONS ERUC/PPI	Audit / supporting field	Local extraction/download date for the ONS source.	0 (0.0%)
eruc_provisional_latest_two_months_flag	ONS ERUC/PPI	Quality / timing control	Marks the latest two ONS observations as potentially provisional.	0 (0.0%)
eruc_latest_12_months_revision_risk_flag	ONS ERUC/PPI	Quality / timing control	Marks recent ONS observations that may be revised.	0 (0.0%)
weight_estimated_by_un_flag	UN Comtrade	Quality / timing control	Supporting provenance, preparation, quality-control or audit field retained in the V6 snapshot.	0 (0.0%)
price_volatility_6m	Derived from trade/FX	Model input	Rolling 6-month volatility of price changes.	6 (1.5%)
quantity_volatility_6m	Derived from trade/FX	Model input	Rolling 6-month volatility of quantity changes.	6 (1.5%)
price_volatility_12m	Derived from trade/FX	Model input	Rolling 12-month volatility of price changes.	12 (3.0%)
quantity_volatility_12m	Derived from trade/FX	Model input	Rolling 12-month volatility of quantity changes.	12 (3.0%)
fx_volatility_6m	Bank of England	Model input	Rolling 6-month volatility of fx changes.	6 (1.5%)
eruc_volatility_6m	ONS ERUC/PPI	Model input	Rolling 6-month volatility of eruc changes.	4 (1.0%)
log_net_weight_kg	Derived feature	Model input	Natural logarithm of monthly import weight, reducing scale skew.	0 (0.0%)
log_partner_count	Derived feature	Model input	Natural logarithm of active supplier-country count.	0 (0.0%)
volume_ratio_to_prior_12m_median	Derived feature	Model input	Current import weight divided by the previous 12-month median.	12 (3.0%)
low_volume_history_flag	Derived feature	Model input	Flags months with unusually low volume and less reliable unit values.	0 (0.0%)
price_shock_history_flag	Derived from trade/FX	Model input	Flags a recent price shock in the observable history.	0 (0.0%)
training_sample_weight	UN Comtrade	Audit / supporting field	Supporting provenance, preparation, quality-control or audit field retained in the V6 snapshot.	0 (0.0%)
target_reliability_score	Derived from trade/FX	Audit / supporting field	Supporting provenance, preparation, quality-control or audit field retained in the V6 snapshot.	0 (0.0%)
target_quality_status	Derived control	Quality / timing control	Supporting provenance, preparation, quality-control or audit field retained in the V6 snapshot.	0 (0.0%)
training_sample_weight_v3	UN Comtrade	Audit / supporting field	Supporting provenance, preparation, quality-control or audit field retained in the V6 snapshot.	0 (0.0%)
supplier_concentration_change_3m	Derived from UN Comtrade	Model input	Three-month change in supplier concentration.	6 (1.5%)
top_supplier_share_change_3m	Derived from UN Comtrade	Model input	Three-month change in the largest supplier's share.	6 (1.5%)

Appendix B. Source file record

Record	Location
Dataset	C:\Anitha\SupplyChainEfficiency\NEW SCI PREDICTION TITANIUM\TRAIN READY Processed data- Titanium 2010-2026\training titanium model A\production_v6_titanium_core_corrected\data\titanium_model_a_production_v6_titanium_core_corrected_training.csv
UN Comtrade source location	C:\Anitha\SupplyChainEfficiency\NEW SCI PREDICTION TITANIUM\Titanium UK monthly imports data 2010-2026\UN ComTrade
Bank of England source	C:\Anitha\SupplyChainEfficiency\NEW SCI PREDICTION TITANIUM\Titanium UK monthly imports data 2010-2026\Bank of Englad database GBP to USD\Bank of England Database.csv
ONS source	C:\Anitha\SupplyChainEfficiency\NEW SCI PREDICTION TITANIUM\Titanium UK monthly imports data 2010-2026\Inflation Data\ppi.csv
Model manifest	C:\Anitha\SupplyChainEfficiency\NEW SCI PREDICTION TITANIUM\TRAIN READY Processed data- Titanium 2010-2026\training titanium model A\production_v6_titanium_core_corrected\results\latest\model_manifest.json
Training code	C:\Anitha\SupplyChainEfficiency\NEW SCI PREDICTION TITANIUM\TRAIN READY Processed data- Titanium 2010-2026\training titanium model A\production_v6_titanium_core_corrected\code\train_model_a_production_v6_titanium_core_corrected.py

Source institutions

UN Comtrade supplies UK import value, weight and partner-country records. The Bank of England supplies the monthly USD-per-GBP conversion series. The UK Office for National Statistics supplies ERUC, the C24 basic-metals import-price index. All model-ready lags, volatility measures, concentration measures, targets and controls are derived locally by the documented preparation and training pipelines.

Record status

This document describes the selected V6 research package and the exact CSV present when the report was generated. Any future refresh should regenerate this PDF so row counts, date coverage, missingness and the manifest feature set remain synchronized.